

# Nico Baier, Ph.D.

734-660-6123 • nico.baier@gmail.com • nicobaier.com •  nbbaier •  nicobaier

## Summary

---

Full-stack TypeScript engineer building AI-powered developer tools and edge-first architectures. Created autonomous AI agents, per-user isolation systems on Cloudflare Durable Objects, and web apps used by 200+ developers. Deep expertise in Cloudflare Workers, React/Next.js, and LLM integration. Track record of rapid prototyping and end-to-end ownership.

## Technical Skills

---

- **Core:** TypeScript, JavaScript, React, Next.js, Node.js, Python, SQL, Git, API Development
- **Edge/Cloud:** Cloudflare Workers, Durable Objects, KV, D1, Vercel, Hono, Alchemy, Val Town
- **AI/LLM:** OpenAI API, Anthropic API, Vercel AI SDK, Model Context Protocol (MCP)
- **Data:** SQLite, PostgreSQL, Turso/LibSQL, PGlite, Drizzle ORM

## Software Projects

---

### Attio List Slack Notifications | Slack Integration | Project [↗](#)

- Built template that ingests Attio CRM webhook events and delivers formatted Slack notifications for list entry and comment activity
- Implemented state-diffing architecture using SQLite to track consecutive snapshots of Attio list entries, enabling change detection without storing computed diffs
- **Tech stack:** TypeScript, Hono, Val Town, SQLite, Attio REST API + Webhooks, Slack Incoming Webhooks

### Idea Explorer | AI Research Agent | GH Repo [↗](#)

- Built an autonomous research assistant using LLMs to explore and refine ideas with tool chaining
- Implemented persistence layer and practical agent workflows for iterative research sessions
- **Tech stack:** TypeScript, Cloudflare Workers, AI SDK, Hono

### Per-User Durable Objects API | Edge Architecture | GH Repo [↗](#)

- Built per-user data isolation architecture using Cloudflare Durable Objects with SQLite storage
- Implemented infrastructure-as-code deployment with Alchemy; each user gets dedicated edge database
- **Tech stack:** TypeScript, Cloudflare Workers, Durable Objects, Drizzle ORM, Hono, Alchemy

### SQLite Explorer | Web App | Project [↗](#)

- Built a SQLite database browser on Val Town used by 200+ developers; rewrote from HTMX to React
- Featured on *Learn with Jason* [↗](#) by Val Town creator Steve Krouse
- **Tech stack:** React, TypeScript, CodeMirror, Hono, Tailwind CSS, Val Town

## HN Notifier | Edge API | GH Repo [↗](#)

- Developed a Cloudflare Workers API for tracking new replies to Hacker News threads
- Implemented endpoints with Zod validation for following threads and managing reply counts
- **Tech stack:** TypeScript, Bun, Hono, Zod, Cloudflare Workers, Cloudflare KV

## Contract Work

---

### Val Town | Freelance Software Engineer | Project [↗](#)

2025

- Built Attio-to-Slack integration template enabling CRM event notifications via webhooks
- Implemented SQLite state management and published as remixable template
- Featured as an official Val Town template [↗](#)
- **Tech stack:** TypeScript, Val Town, Hono, SQLite, Slack Webhooks, Attio API

### Mastra | Contributor | Case Studies [↗](#)

2025

- Authored technical case studies documenting enterprise AI workflow implementations
- Conducted customer interviews to capture integration patterns and developer adoption stories

## Professional Experience

---

### Natural Language Analyst

Feb 2023 – Jan 2025

Bold, Ltd.

- Built and maintained a library of 15+ SPARQL queries for common data operations, increasing efficiency by 83%
- Implemented data analysis workflows for English, German, Dutch, and Polish pipelines
- Led a team of 7 data contractors, ensuring data quality across 5 language pipelines

### Inbound Specialist

Apr 2021 – May 2022

Stripe, Inc.

- Wrote R scripts to standardize linguistic data from applicant tracking systems, reducing errors by 22%
- Established data-driven processes for new inbound sourcing department, leading to 30% faster resume review

### Postdoctoral Researcher & Lecturer

Sep 2017 – Jul 2020

University of British Columbia & McGill University

- Designed metadata standards for a 12,500+ entry text dataset; managed team of 15 for data collection and QA

## Education

---

- **Ph.D., Linguistics** • 2018, University of California, Berkeley
- **M.A., Linguistics** • 2013, University of California, Berkeley
- **B.A., Linguistics & German** • 2010, University of Michigan

## Languages

---

- **English** (Native)
- **German** (Fluent)